Euclid's Elements — Book XI
Proposition 14

Formal Reconstruction — Technical Documentation

Jurek Różański
Military University of Technology, Warsaw, Poland
jerzy.rozanski@wat.edu.pl

CGJteamLab

Planes to which the same straight line is at right angles are parallel.

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1 Introduction

Proposition XI.14 states that planes to which the same straight line is at right angles are parallel. The classical argument is short: if the two planes met, a point of their intersection together with the common perpendicular would produce a triangle with two right angles, contradicting Proposition I.17.

The formal reconstruction preserves this geometric core, but it exposes two points that the classical presentation leaves implicit.

First, Euclid's contradiction argument requires the two feet of the common perpendicular to be distinct. If the line meets both planes at the same point, the triangle used in the proof degenerates. The production proof therefore separates an equal-foot uniqueness theorem from the normalized case $A \neq B$.

Second, Euclid reaches a common point by first speaking about the line of intersection of the two planes. In the formal library, parallelism of two planes is defined directly as absence of a common point. Negating the target therefore gives the point K immediately; the proof does not need to construct the whole intersection line before beginning the contradiction.

These two differences make XI.14 a useful example of the project's general method. The classical geometry is retained, while diagrammatic assumptions and representational shortcuts are isolated as explicit proof obligations.

2 Classical Configuration

Let l be a straight line perpendicular to plane π at A and to plane ρ at B:

$$l \perp \pi \text{ at } A, \qquad l \perp \rho \text{ at } B.$$

The intended nontrivial configuration has

$$\pi \neq \rho, \qquad A \neq B.$$

Euclid's conclusion is that the two planes do not meet.

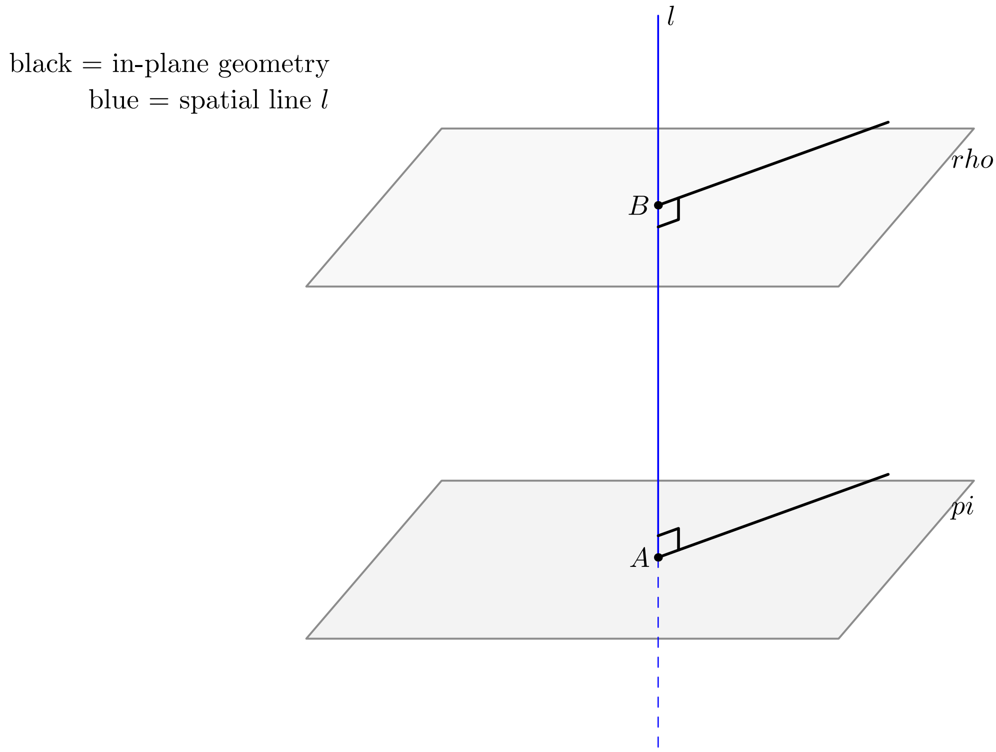


Figure 1: The basic spatial configuration of Proposition XI.14. The common perpendicular l meets the two reference planes at A and B. Following the Book XI drawing convention, geometry internal to a displayed plane is black, while genuinely spatial objects are blue.

The formal public theorem also carries the hypothesis $\pi \neq \rho$. This matches the nontrivial reading of Euclid XI.Def.8 used in the project: parallel planes are distinct planes having no common point.

Diagrammatic audit. Three stable geometric states are sufficient for the documentation of XI.14. Figure 1 records the input configuration. Figure 3 records the hypothetical common-point state and the reduction to a planar triangle. Figure 2 records the exceptional equal-foot uniqueness argument. No additional figure is needed for the internal SameRay/opposite-ray case split, because that split is a formal orientation normalization rather than a new stable Euclidean construction.

3 Statement

Theorem 1 (Euclid XI.14)**.** *Let π and ρ be distinct planes, and let l be a straight line. If l is perpendicular to π at A and perpendicular to ρ at B, then π and ρ are parallel.*

The current production declaration is:

```
theorem euclid_proposition_11_14
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
```

```
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [_HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (pi rho : S.Plane)
    (l : Geo.Line)
    (A B : Geo.Point)
    (hPlanesNe : Ne pi rho)
    (hPerpPi :
      HilbertLinePerpendicularPlaneAt Geo l pi A)
    (hPerpRho :
      HilbertLinePerpendicularPlaneAt Geo l rho B) :
    HilbertSpacePlanesParallel Geo pi rho
```

The target predicate is proposition-local:

```
def HilbertSpacePlanesParallel
    [S : HilbertSpacePrimitive Geo]
    (pi rho : S.Plane) : Prop :=
  Not
    (exists X : Geo.Point,
      S.OnPlane X pi /\
      S.OnPlane X rho)
```

Thus the conclusion is represented directly as nonintersection of the two planes.

4 Classical Proof

Assume, for contradiction, that the two planes meet. By XI.3 their common section is a straight line; choose a point K on this section.

Join A to K and B to K. Since $K \in \pi$, the line AK lies in π. Because $l \perp \pi$ at A, Definition XI.3 gives

$$l \perp AK.$$

Similarly, since $K \in \rho$,

$$l \perp BK.$$

The lines l and the point K determine a plane σ. The points A and B lie on l, so the triangle ABK lies in σ. Inside this plane,

$$\angle BAK \text{ is right}, \qquad \angle ABK \text{ is right}.$$

But I.17 says that two angles of a triangle are together less than two right angles. A nondegenerate triangle therefore cannot have two right angles. This contradiction proves that π and ρ do not meet.

The local classical dependency pattern is

$$\begin{array}{c} l \perp \pi, \quad l \perp \rho \\ \downarrow \\ \text{assume } \pi \cap \rho \neq \varnothing \\ \downarrow \\ \text{XI.3} \\ \downarrow \\ K \in \pi \cap \rho \\ \downarrow \\ \text{XI.Def.3} \\ \downarrow \\ \angle BAK, \angle ABK \text{ right} \\ \downarrow \\ \text{I.17} \\ \downarrow \\ \perp. \end{array}$$

The proof is neutral. No Euclidean parallel postulate is involved.

5 The Equal-Foot Issue

The classical triangle argument silently presupposes $A \neq B$. If $A = B$, the side AB collapses and the I.17 contradiction cannot even be formed.

The production proof handles this branch by proving a separate uniqueness theorem: for a fixed line l and a fixed point A, there is at most one plane perpendicular to l at A.

The construction is synthetic. Choose two distinct lines m, n through A in π. Since $l \perp \pi$, both are perpendicular to l. Now let p be any line through A in ρ. Since $l \perp \rho$, the line p is also perpendicular to l. Proposition XI.5 then places m, n, p in one plane. Because the two distinct intersecting lines m, n already determine π, the line p must lie in π.

Applying the same argument to two distinct lines through A in ρ shows that two distinct intersecting lines of ρ also lie in π. The uniqueness of the plane through two intersecting lines yields

$$\rho = \pi.$$

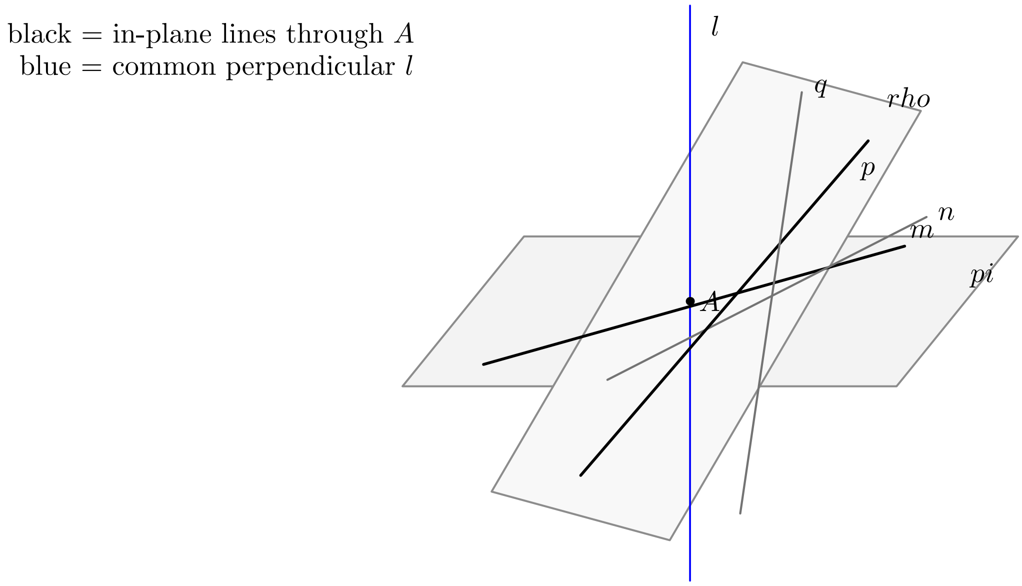
This is not an auxiliary side remark. It is exactly what allows the public theorem to derive $A \neq B$ from the assumed plane distinctness $\pi \neq \rho$.

6 What is genuinely three-dimensional here?

XI.14 has a particularly clear spatial-to-planar transition.

The genuinely three-dimensional data are:

- two ambient planes π, ρ;
- a single ambient line l perpendicular to each;



equal-foot uniqueness case

Figure 2: The exceptional case in which the same line l is perpendicular to both planes at the same point A. Two distinct in-plane lines through A, together with XI.5, force every line through A in the second plane into the first plane; hence the two planes coincide.

- a hypothetical point K common to the two planes;
- the auxiliary plane σ through l and the off-line point K;
- the two connector lines AK and BK, each simultaneously controlled by one reference plane and by σ.

Once σ has been constructed, the terminal contradiction is planar. The proof passes to

$$\text{PlaneGeo}(\sigma)$$

and works with the induced Hilbert incidence, order, and congruence structure. The two spatial perpendicularities are converted into the exact right-angle statements for the triangle ABK, after which I.17 is purely two-dimensional.

Thus XI.14 is not merely a planar theorem drawn in space. The proof must first manufacture the correct plane slice and transport the ambient line–plane perpendicularity data into it.

7 Spatial / Planar Architecture Used

The ambient layer uses

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertLineInPlane
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
```

The key spatial incidence constructors are

```
hilbert_plane_through_line_and_external_point
hilbert_plane_through_two_intersecting_lines
```

For the normalized branch, the proof constructs σ from the common normal l and the hypothetical common point K. The points and lines are then re-packaged as

```
PlanePoint Geo sigma
PlaneLine Geo sigma
PlaneGeo Geo sigma
```

This passage is essential because Proposition I.17 is a planar theorem. The bridge layer transfers incidence and perpendicularity from the ambient spatial carriers to the induced plane geometry.

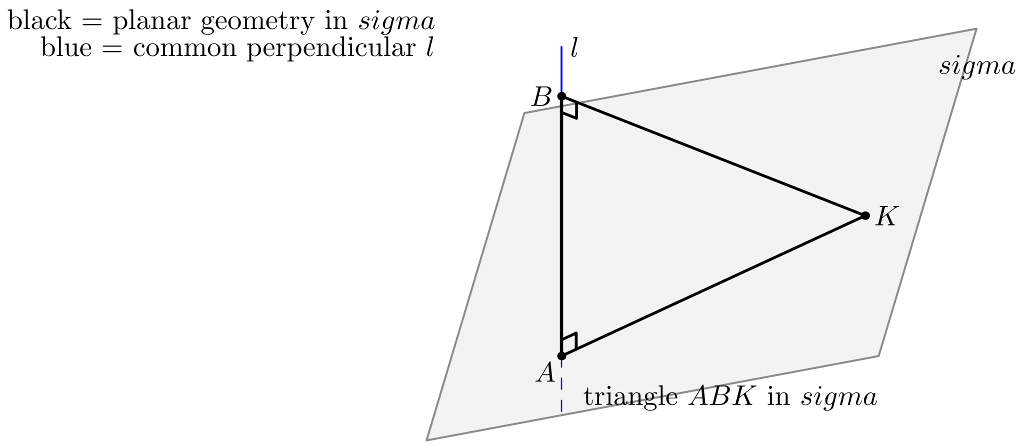


Figure 3: The hypothetical common point K and the common perpendicular l determine the auxiliary plane σ. In σ, the triangle ABK has right angles at both A and B. The spatial construction is therefore reduced to the planar contradiction supplied by I.17.

8 Proof Architecture of the Reconstruction

8.1 1. Excluding a common point on the normal

Assume $A \neq B$ and suppose K lies on both planes. If K also lay on l, the perpendicular-foot uniqueness theorem from the XI.12 infrastructure would give

$$K = A \qquad \text{and} \qquad K = B,$$

contradicting $A \neq B$.

This is formalized by `hilbert_XI14_common_point_off_normal`.

8.2 2. Constructing the auxiliary plane

Because $K \notin l$, the general spatial incidence theorem constructs a unique plane σ containing l and K. Since $A, B \in l$, all three vertices A, B, K lie in σ.

This stage is the formal replacement for what the classical diagram makes immediate.

8.3 3. Constructing the connector lines

The lines AK and BK are explicitly constructed. The first lies in both π and σ; the second lies in both ρ and σ. Definition XI.3, encoded by `HilbertLinePerpendicularPlaneAt.perpendicular_to_line`, gives

$$l \perp AK \text{ at } A, \qquad l \perp BK \text{ at } B.$$

8.4 4. Passing to `PlaneGeo`

The ambient configuration is transferred to the induced plane geometry of σ. The noncollinearity of A, B, K follows because $A, B \in l$ while $K \notin l$.

At this point the spatial part of the normalized contradiction is complete.

8.5 5. Recovering the exact triangle angles

The formal definition of line–line perpendicularity contains witnesses on the two carrier lines. Those witnesses need not initially be on the exact rays AB, AK and BA, BK used in the triangle.

The production proof therefore performs an orientation normalization. A collinear witness is either on the same ray from the vertex as the desired triangle point or on the opposite ray. SameRay transport and the two opposite-ray right-angle transport lemmas show that rightness is invariant under all four orientation combinations.

The result is the stable geometric statement

$$\angle BAK \text{ right}, \qquad \angle ABK \text{ right}.$$

This is packaged by `hilbert_XI14_triangle_has_two_right_angles`.

8.6 6. Proposition I.17 closes the planar contradiction

Apply the I.17 variant `euclid_proposition_17_BAC_ABC` to the triangle ABK. It extends the side BK through B to a point E and yields

$$\angle BAK < \angle ABE.$$

Since $K - B - E$ and $\angle ABK$ is right, the angle $\angle ABE$ is also right. All right angles are congruent, so

$$\angle BAK \cong \angle ABE.$$

A strict angle inequality cannot hold between congruent angles. This gives the contradiction.

The planar closure is packaged by `hilbert_XI14_two_right_angles_impossible`.

8.7 7. The normalized theorem

The theorem `euclid_proposition_11_14_normalized` now has a very short public shape: assume a common point K, construct the two-right-angle triangle, and invoke the planar impossibility theorem.

8.8 8. Removing the normalization

The public theorem must not assume $A \neq B$. It derives this from $\pi \neq \rho$.

If $A = B$, the equal-foot uniqueness theorem `hilbert_XI14_plane_perpendicular_to_line_at_unique` gives $\pi = \rho$, contradicting the hypothesis. Therefore $A \neq B$, and the normalized theorem applies.

9 Lean Formalization

The production module imports

```
import CGJteamLab.Proposition11_13
import CGJteamLab.Proposition11_5
import CGJteamLab.Proposition17
import CGJteamLab.HilbertRightAngle
```

The direct mathematical engines used by XI.14 are distributed across these imports:

- XI.5 supplies the equal-foot coplanarity argument;
- the XI.12 infrastructure supplies perpendicular-foot uniqueness;
- I.17 supplies the terminal planar angle inequality;
- the Hilbert right-angle library supplies transport and congruence of right angles.

The public proof itself is intentionally small. Almost all of the work is performed in reusable proposition-local lemmas that separate the spatial construction, the planar reduction, the angle-orientation normalization, and the equal-foot uniqueness theorem.

This organization mirrors the mathematical architecture more faithfully than a single monolithic tactic proof would.

10 Hidden Formal Data

10.1 Plane distinctness is part of the public statement

The project defines parallel planes by nonintersection, but XI.14 is intended for two distinct planes. The hypothesis `hPlanesNe` therefore appears explicitly.

It is also what rules out the equal-foot branch.

10.2 The common point must be off the normal

The classical picture places K away from AB automatically. Lean must prove this. The argument uses perpendicular-foot uniqueness in each reference plane.

10.3 The auxiliary plane must be carried explicitly

The sentence "draw the plane through AB and K" hides several facts: $K \notin l$, the existence of the plane, containment of l, and containment of K. All are retained because later bridge lemmas depend on them.

10.4 Perpendicularity witnesses have orientation

A statement such as $l \perp AK$ is not yet syntactically the statement that $\angle BAK$ is right. The witness points appearing in the definition of perpendicular lines may occupy either ray of the carrier. The production proof therefore has a genuine four-case SameRay/opposite-ray normalization.

This is formal data hidden completely by the Euclidean diagram.

10.5 I.17 returns an angle inequality, not the final contradiction

The formal I.17 theorem yields a strict angle comparison after extending one side of the triangle. XI.14 must still prove that the resulting exterior angle is right and then use congruence of all right angles to contradict the strict inequality.

11 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses XI.3 and Definition XI.3.

The formal dependency graph is reorganized.

The content corresponding to XI.3 is supplied more economically by the definition of plane parallelism and by general spatial incidence constructors. Since the negation of `HilbertSpacePlanesParallel` already gives a common point K, the proof does not first need an explicit line representing the complete intersection $\pi \cap \rho$.

Definition XI.3 is represented directly by `HilbertLinePerpendicularPlaneAt`: every line of the reference plane through the foot is perpendicular to the given line.

XI.5 enters only in the equal-foot uniqueness branch. This is a genuine new formal dependency caused by making explicit the case $A = B$.

XI.12 contributes the previously proved uniqueness of the perpendicular foot of a fixed line in a fixed plane. XI.13 itself is not called by the proof; its module is an immediate production import in the Book XI sequence.

12 Relationship with Book I / Book Zero / Hilbert Infrastructure

The decisive Book I dependency is Proposition I.17. In the current library the relevant theorem is `euclid_proposition_17_BAC_ABC`. It supplies the strict comparison needed to rule out the two-right-angle triangle.

The right-angle library contributes two separate ingredients:

- transport of a right angle when one of its arms is replaced by the same or opposite ray;
- congruence of arbitrary right angles.

The spatial proof uses Hilbert incidence, order, and congruence only. There is no `HilbertSpaceEuclideanGeo` parameter in the theorem signature. Thus the reconstruction is neutral with respect to Hilbert Group IV.

No coordinate system, vector space, scalar product, normal vector, numerical angle measure, or trigonometric function occurs anywhere in the proof.

Book Zero contributes only inherited general infrastructure through the existing project layers; XI.14 introduces no new proposition-specific foundational principle.

13 Dependency Summary

The classical architecture is

$$\begin{gathered} l \perp \pi, \quad l \perp \rho \\ \downarrow \\ \text{assume the planes meet} \\ \downarrow \\ \text{XI.3: intersection line} \\ \downarrow \\ K \in \pi \cap \rho \\ \downarrow \\ AK \subset \pi, \quad BK \subset \rho \\ \downarrow \\ \text{XI.Def.3} \\ \downarrow \\ \angle BAK, \angle ABK \text{ right} \\ \downarrow \\ \text{I.17} \\ \downarrow \\ \bot \\ \downarrow \\ \pi \parallel \rho. \end{gathered}$$

The formal architecture has two branches:

$$\begin{array}{c}
\pi \neq \rho, \quad l \perp \pi \text{ at } A, \quad l \perp \rho \text{ at } B \\
\downarrow \\
A = B\ ? \\
\begin{array}{cc}
\swarrow & \searrow \\
A = B & A \neq B \\
\downarrow & \downarrow \\
\text{XI.5 + plane uniqueness} & \text{assume common point } K \\
\downarrow & \downarrow \\
\pi = \rho & K \notin l \\
\downarrow & \downarrow \\
\perp & \sigma = \text{plane}(l, K) \\
 & \downarrow \\
 & \text{PlaneGeo}(\sigma) \\
 & \downarrow \\
 & \angle BAK, \angle ABK \text{ right} \\
 & \downarrow \\
 & \text{I.17} \\
 & \downarrow \\
 & \perp
\end{array} \\
\downarrow \\
\text{no common point of } \pi, \rho.
\end{array}$$

The main conceptual difference is therefore clear. Euclid has one visible contradiction branch and suppresses the degenerate-foot possibility. The formal proof makes that possibility explicit and discharges it by a uniqueness theorem before entering the classical branch.

14 Audit Status

This documentation pass was checked against the current production `Proposition11_14.lean` and the Book XI source package through Proposition XI.14. The production theorem was reported to compile successfully, and the separate audit file was reported to compile without introducing a new proposition-specific geometric axiom.

The documentation deliberately does not reproduce the raw `#print axioms` output as a separate mathematical section. The relevant structural fact is recorded above: the theorem has no Group IV parameter and uses only the already accepted project infrastructure.

The Asymptote figures follow the Book XI convention exemplified by the arbitrary-transversal figure for XI.4: geometry internal to a displayed plane is black, genuinely spatial objects are blue, hidden spatial segments are blue dashed, and plane fills are kept light.

15 Conclusion

Proposition XI.14 has a simple classical idea but a rich formal anatomy.

The main Euclidean argument survives unchanged. A hypothetical common point of the two planes determines, together with the common perpendicular, an auxiliary plane. In that plane one obtains a nondegenerate triangle with two right angles, and Proposition I.17 gives the contradiction.

The formal reconstruction adds exactly what the classical diagram suppresses. It proves that the hypothetical common point is off the normal, constructs and retains the auxiliary plane explicitly, normalizes the orientation of the perpendicularity witnesses, and separates the degenerate equal-foot case. Proposition XI.5 then supplies a synthetic uniqueness theorem showing that two planes perpendicular to the same line at the same point must coincide.

The result is therefore both faithful to Euclid's spatial proof and sharper about its hidden assumptions. It also illustrates a recurring Book XI pattern: the genuinely spatial work consists in constructing and controlling the correct carrier plane; once that plane is available, the terminal argument returns to the planar Hilbert geometry already developed for Book I.